

Unit 1

Set Theory

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1.1 Introduction

The concept of sets is fundamental to all branches of mathematics. (The creation of set theory is due to the German mathematician George Cantor (1815-1998). He was born on March 3rd, 1815 at Petersburg in Russia.) Consider any dictionary of English. It is a collection of words and their meanings. A word either belongs to this collection or not, depending on whether it is listed in the dictionary or not. This collection is an example of a set.

The study of sets includes the study of operations on sets. In this unit we discuss the operations of complementation, union, intersection and Cartesian product. We also introduce Venn diagrams, a pictorial way of describing sets.

Since the material covered in this unit is going to be basis for the rest of the course, please study it carefully.

Objectives

After studying this unit you should be able to

- Identify and describe a set
- Perform the operation of complementation, union and intersection on sets

1.2 Sets

A set is a collection of well-defined objects. The term well-defined means that with the help of the given rule, it is possible to tell whether a given object belongs to or does not belong to the given set. For example, the collection of all chemical elements is a set. This is because any object is either a chemical element or not, and accordingly it does or does not belong to the collection. On the other hand, the collection of all intelligent human beings is not a set. Why ? Because, given a human being, one person may feel that she or he is intelligent while another may not. So, the collection is not well-defined.

Now, we give some more examples of sets which we often come across.

- i. The set of all natural numbers denoted by N
- ii. The set of all integers denoted by Z
- iii. The set of all rational numbers, denoted by Q
- iv. The set of all real numbers, denoted by R
- v. The set of all organic compounds

The objects of the set are called the elements or members of the set. Sets are generally denoted by capital letters, A, B, C, X, Y, Z etc., and their elements by a, b, cetc. If an object 'a' is a member of a set A , then we write $a \in A$ and if it is not a member, then we write $a \notin A$. There are two methods of representing a set.

1.2.1 Tabular (or Roster Method): Here we list out all its members and write them within braces $\{ \}$

For example, the set V of vowels in English alphabet is written in this method as $V = \{a, e, i, o, u\}$

The set of natural numbers less than 5 is $A = \{1, 2, 3, 4\}$

1.2.2 Rule (or set builder) Method: Here a property satisfied by every element of the set is stated. For example

$$A = \{x \mid x \text{ is an vowel of English alphabet}\}$$

$$B = \{x \mid x \text{ is a natural number } < 5\}$$

1.3 Types of Sets

- i) **Null set (empty or void set):** It is a set containing no elements. Null set is denoted by ϕ or $\{ \}$

For example

- 1) The set of integers satisfying the equation $3x + 1 = 0$ is a null set.
- 2) The set of real numbers satisfying $x^2 + 1 = 0$ is a null set.

- ii) **Equal sets:** Two sets A and B are said to be equal, written $A=B$, if every element of A is in B and every element of B is in A .

$$\text{e.g. } A = \{1, 2, 3\}, B = \{2, 3, 1\} \text{ then } A = B$$

$$A = \{1, 2, 3\}, B = \{1, 2, 3, 4\} \text{ then } A \neq B$$

- iii) **Singleton set:** A set consisting of a single element is called a singleton sets or simply a singleton.

e.g. $\{5\}$ is a singleton set.

$\{ \}$ is not a singleton but a null set.

- iv) **Non-empty set:** A set which is not empty is a non-empty set. That is, it consists atleast one element.

- v) **Finite and infinite sets:** If the number of elements of a set is finite, then it is called a finite set, otherwise it is called an infinite set.

- vi) **Cardinal number of a finite set:** It is the number of distinct elements of the set A and this number is denoted by $n(A)$.

Eg. $A = \{1, 2, 3\}$ is a finite set $n(A) = 3$

$N = \{1, 2, 3, 4, \dots\}$ is an infinite set.

- vii) **Subsets:** Let A and B be two sets such that if every element of A is in B then A is said to be a subsets of B and this is denoted by $A \subseteq B$.

If A is a subset of B , then B is called a superset of A and is written $B \supseteq A$.

Eg. $A = \{1, 2, 3, 4\}$ $B = \{1, 2, 3, 4, 5\}$

- viii) **Proper subsets:** If $A \subseteq B$ and $A \neq B$ then A is said to be a proper subset of B and this is written $A \subset B$.

In this case, there is atleast one element of B which is not in A .

Eg. $A = \{a, b, c\}$ $B = \{a, b, c, d\}$

- ix) **Universal set:** In a mathematical discussion of sets we usually consider all the sets to be subsets of a set U , called universal set.

Eg. $A = \{1, 2\}$, $B = \{3, 4, 5\}$, $C = \{2, 5, 6\}$ then $U = \{1, 2, 3, 4, 5, 6\}$

Note:

- i. Every set is a subset of itself
- ii. Null set $\emptyset$ is the subset of every set
- iii. Universal set is the super set of all sets under discussion.

- x) **Power set:** The family or collection of all subsets of a given set A is called the power set of A and it is denoted by $P(A)$.

Eg. $A = \{1, 2, 3\}$

$P(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$

Note: If $n(A) = p$ then $n[P(A)] = 2^p$

Compliment of set A: It is denoted by A^c or A^1

or A and it is the set of elements of the universal set U which are not in A

Eg. $A = \{1, 2\}$, $B = \{2, 3, 4\}$ and $U = \{1, 2, 3, 4\}$ $A^c = \{3, 4\}$ and $B^c = \{1\}$

1.4 Operation on Sets

Definition: Union of two sets A and B is the set of all elements which are in A or B or in both and it is denoted by $A \cup B$ (read as, A union B or A cup B).

Symbolically $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Eg. $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$ then $A \cup B = \{1, 2, 3, 4, 5\}$

Definition: Intersection of two sets A and B is the set of elements which are both in A and B and this is denoted by $A \cap B$ (read as A intersection B or A cap B)

Eg. $A = \{a, b, c\}$, $B = \{b, c, d, e\}$ then $A \cap B = \{b, c\}$

Note: If $A \cap B = \emptyset$ then A and B are said to be disjoint

Difference ($A - B$) of two sets A and B is set of elements A that are not in B

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

Eg. $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5\}$, then $A - B = \{1\}$.

Definition: Symmetric difference of two sets A and B is denoted by $A \Delta B = (A - B) \cup (B - A)$

Eg. $A = \{1, 2, 3\}$ and $B = \{2, 3, 4, 5\}$ then $A - B = \{1\}$, $B - A = \{4, 5\}$

$$A \Delta B = (A - B) \cup (B - A) = \{1, 4, 5\}$$

1.4.1 Ordered Pairs: If a and b are any two elements, then (a, b) is an ordered pair in which ' a ' is the first element and ' b ' is the second element.

In general $(a, b) \neq (b, a)$ when $a \neq b$

If $(a, b) = (c, d)$ then $a = c$ and $b = d$.

Eg. If $(x, x + y) = (3, 5)$ then $x = 3$, and $x + y = 5 \implies y = 2$

1.5 Venn Diagrams

Sets and their relationship can be represented by diagrams, called Venn diagrams. An English logician John Venn (1834 –1923), invented this. In

these diagrams sets of any kind are graphically represented by sets of points.

Consider three sets A , B and C . We choose universal set U . The Venn diagram of such a situation would be drawn as follows. The rectangle represents U . The subsets A , B and C are represented by closed regions lying completely within the rectangle. These regions may be in the form of a circle, ellipse or any other shape. To clarify what we just said, consider the following example.

1.5.1 Illustration on Venn diagrams

Draw a Venn diagram to represent the sets

$$U = \{1, 2, \dots, 10\}, A = \{1, 2, 3\}, B = \{3, 4, 5\}, C = \{6, 7\}$$

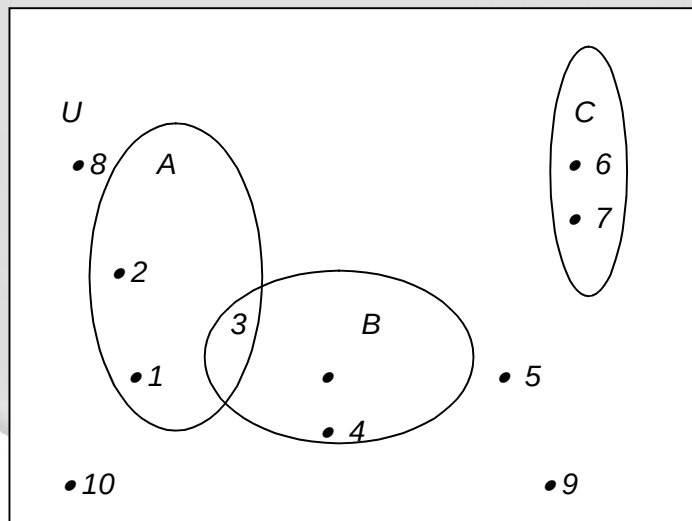


Fig. 1

The points 8, 9, 10 do not lie in any of A , B or C as the same can be observed from the Figure.

Note that 3 belongs to both A and B . It lies in the intersected area of A & B . Also note that A and C do not have any element in common. Therefore, the

regions representing them do not cut each other. For the same reason the regions representing B and C do not cut each other. Ofcourse, we could have drawn all three sets in any one shape.

What happens, if $A \subset B$, that is A is a proper subsets of B . Well then, we can just take B be our universal set.

The Venn diagram in Fig. 2 represents this situation.

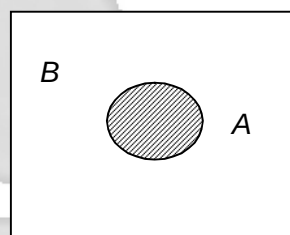


Fig. 2

Now let us represent the intersection of sets by means of Venn diagrams.

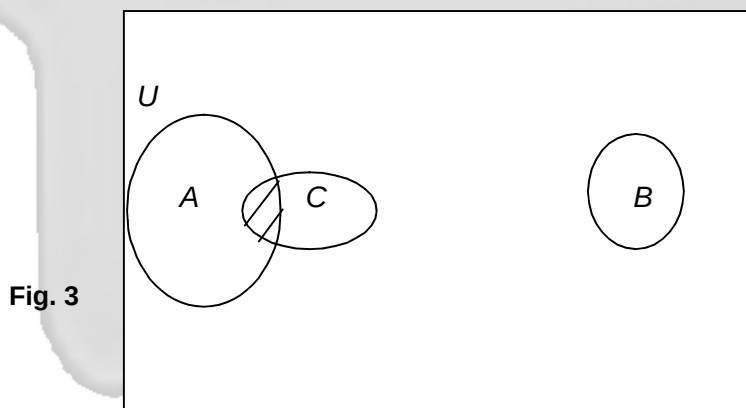


Fig. 3

A and B have no common element, this is $A \cap B = \emptyset$, from the diagram.

The shaded region in Fig. 3 represents the set $A \cap C$. It can also be seen that neither $A \subseteq C$ nor $C \subseteq A$. Both $C - A$ and $A - C$ are non empty sets.

What situation does Fig. 4 represent ? It shows two sets A and B with $A \subset B$, that is, A is a proper

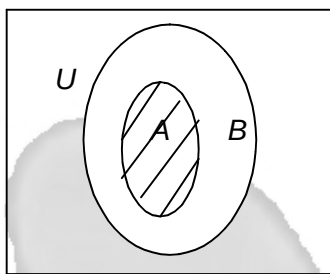


Fig. 4

subset of B . Then the shaded area shows $A \cap B = A$.

It is easy to visualise union of sets by Venn diagrams. Consider Fig. 5. In this diagram we see

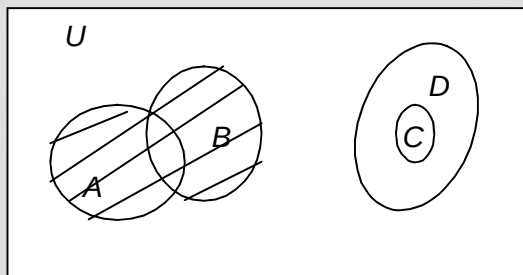


Fig. 5

four sets A , B , C and D and the universal set U . The shaded area represents $A \cup B$. This area, along with the area enclosed by D , represents $A \cup B \cup D$. $C \cup D$ is the area enclosed by C and D , which is just D , since $C \subseteq D$.

1.6 Cartesian Product of two Sets

If A and B are any two non-empty sets, then the Cartesian product of A and B is denoted by $A \times B$ and it is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

$$\text{i.e. } A \times B = \{(a, b) \mid a \in A, b \in B\}$$

eg. $A = \{1, 2, 3\}$, $B = \{a, b\}$ then

$$A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$$

$$B \times A = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

$$B \times B = \{(a, a), (a, b), (b, a), (b, b)\}$$

$$A \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

Note:

- i. In general $A \times B \neq B \times A$
- ii. If A or B is a null set, then $A \times B = \emptyset$
- iii. If $n(A) = p$, $n(B) = q$ then $n(A \times B) = pq$

Worked examples:

1. Find x and y if $(3x + y, x - 1) = (x + 3, 2y - 2x)$

Solution: The ordered Pairs are equal if $3x + y = x + 3$ and $x - 1 = 2y - 2x$

$$\text{i.e. } 2x + y = 3$$

$$3x - 2y = 1$$

Solving $x=1, y=1$

2. If $A = \{1, 2, 3\}$ $B = \{2, 4, 5\}$ find

$$\text{i. } (A \cap B) \times (A - B)$$

$$\text{ii. } A \times (A - B)$$

$$\text{iii. } (A \Delta B) \times (A \cap B)$$

Now,

$$A \cap B = \{2\}$$

$$A - B = \{1, 3\}$$

$$B - A = \{4, 5\}$$

$$A \Delta B = (A - B) \cup (B - A) = \{1, 3, 4, 5\}$$

$$\text{i. } (A \cap B) \times (A - B) = (2) \times \{1, 3\} = \{(2, 1), (2, 3)\}$$

$$\text{ii. } A \times (A - B) = \{1, 2, 3\} \times (1, 3) = \{(1, 1), (1, 3), (2, 1), (2, 3), (3, 1), (3, 3)\}$$

$$\text{iii. } (A \Delta B) \times (A \cap B) = \{1, 3, 4, 5\} \times \{2\} = \{(1, 2), (3, 2), (4, 2), (5, 2)\}$$

- 3) If $A = \{x/x \in N \text{ and } x < 3\}$, $B = \{x/x^2 - 16 = 0 \text{ and } x < 0\}$ find $B \times A$ where N is a set of natural number

Solution:

$$\begin{array}{l|l} A = \{1, 2\} \text{ and } B = \{-4\} & \because x^2 - 16 = 0 \\ & (x - 4)(x + 4) = 0 \\ B \times A = \{-4\} \times \{1, 2\} & \\ = \{(-4, 1), (-4, 2)\} & \implies x = -4, 4 \\ & \therefore x = -4 (\because x < 0) \end{array}$$

- 4) If $A = \{x \mid x \text{ is a positive prime. There are no negative primes } < 8\}$ $B = \{6, 7, 8\}$, $C = \{7, 8, 9\}$

find $(A \cap B) \times (B \cap C)$

A prime number is a natural number other than one whose only factors are one and itself

Solution: $A = \{2, 3, 5, 7\}$ $B = \{6, 7, 8\}$ $C = \{7, 8, 9\}$

$$(A \cap B) = \{7\} \text{ and } (B \cap C) = \{7, 8\}$$

$$(A \cap B) \times (B \cap C) = \{7\} \times \{7, 8\} = \{(7, 7), (7, 8)\}$$

- 5) If $A = \{x/x^2 - 5x + 6 = 0\}$ $B = \{2, 4\}$ $C = \{4, 5\}$ find $(A - B) \times (B - C)$

Solution:

$$x^2 - 5x + 6 = 0 \implies (x - 2)(x - 3) = 0 \implies x = 2, 3$$

$$A = \{2, 3\}, B = \{2, 4\} \text{ and } C = \{4, 5\}$$

$$A - B = \{3\} \text{ and } B - C = \{2\}$$

$$\text{Therefore } (A - B) \times (B - C) = (3) \times (2) = (3, 2)$$

6) Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$ and $C = \{1, 4, 7, 8\}$ determine

$A \cap B \cap C = (A \cap B) \cap C$ Also verify that

$$a) A \cap B \cap C = (A \cap B) \cap C$$

$$b) A \cap B \cap C = A \cap (B \cap C)$$

$$(A \cap B) = \{3, 4\}, (B \cap C) = \{4\}$$

$$(A \cap B) \cap C = \{3, 4\} \cap \{1, 4, 7, 8\} = \{4\}$$

$$A \cap (B \cap C) = \{1, 2, 3, 4\} \cap \{4\} = \{4\}$$

$$\text{Therefore } A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C)$$

1) Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 5, 6\}$ verify that $A \cup B = (A - B) \cup$

$$(A \cap B) \cup (B - A)$$

$$A \cup B = \{1, 2, 3, 5, 6\}$$

$$A - B = \{1\}, (A \cap B) = \{2, 3\}$$

$$B - A = \{5, 6\}$$

$$(A - B) \cup (A \cap B) \cup (B - A) = \{1, 2, 3, 5, 6\}$$

$$\text{Therefore } (A \cup B) = (A - B) \cup (A \cap B) \cup (B - A)$$

1.7 Distributive Laws, De Morgan's Law

You must be familiar with the distributive law that connects the operations of multiplication and addition of real numbers. It is

$$a \times (b + c) = (a \times b) + (a \times c), a, b, c \in R$$

Similarly, we have two distributive laws that relate the union and intersection of sets. They are

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \dots\dots\dots (1)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \dots\dots\dots (2)$$

for any three sets A , B and C . In (1) $\cap$ is distributive over U and in (2) U is distributive on $\cap$. We will not prove them but will illustrate their validity by some examples and exercises.

Eg. Let $A = \{1\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$

Verify that

$$a. A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$b. A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(B \cup C) = \{2, 3, 4, 5\}$$

$$A \cap (B \cup C) = \{1\} \cap \{2, 3, 4, 5\} = \emptyset \dots\dots\dots (1)$$

$$(A \cap B) = \emptyset, (A \cap C) = \emptyset$$

$$(A \cap B) \cup (A \cap C) = \emptyset \dots\dots\dots (2)$$

$$\text{From (1) and (2) } A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$B \cap C = \{2, 3, 4\} \cap \{3, 4, 5\} = \{3, 4\}$$

$$A \cup (B \cap C) = \{1\} \cup \{3, 4\} = \{1, 3, 4\} \dots\dots\dots (3)$$

$$A \cup B = \{1, 2, 3, 4\}$$

$$A \cup C = \{1, 3, 4, 5\}$$

$$(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4\} \cap \{1, 3, 4, 5\} = \{1, 3, 4\} \dots\dots\dots (4)$$

From (3) and (4)

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Now we state two more laws that relate the operation of finding the complement of a set to that of the intersection or union of sets. These are known as DeMorgan's laws, after the British mathematician Augustus De Morgan (1806– 1871).

They state that

$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

for any two subsets A and B of a universal set U .

Let us verify these laws for the following example

Eg.1. Let $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$ and $B = \{2, 3, 4\}$

Verify De Morgan's Laws

Solution :

$$A^c = \{3, 4\}, B^c = \{1\}$$

$$\text{So that } A^c \cup B^c = \{1, 3, 4\} \dots\dots\dots(1)$$

$$\text{Also } A \cap B = \{2\}$$

$$(A \cap B)^c = \{1, 3, 4\} \dots\dots\dots(2)$$

From (1) and (2)

$$(A \cap B)^c = A^c \cup B^c$$

$$\text{Now, } (A \cup B) = \{1, 2, 3, 4\} = U$$

$$(A \cup B)^c = \emptyset \dots\dots\dots(3)$$

$$A^c \cap B^c = \emptyset \dots\dots\dots(4)$$

From (3) and (4)

$$(A \cup B)^c = A^c \cap B^c$$

Eg. 2: $U = \{1, 2, 3, 4, \dots\dots\dots\}$, $A = \{2, 4, 6, \dots\dots\dots\}$ and $B = \{1, 3, 5, \dots\dots\dots\}$

verify DeMorgan's Laws

Solution:

$$A^c = \{1, 3, 5, \dots\dots\dots\} = B$$

$$B^c = \{2, 4, 6, \dots\dots\dots\} = A$$

So that

$$A^c \cup B^c = B \cup A = U$$

$$(A \cap B)^c = \emptyset \therefore (A \cap B)^c = U$$

$$\text{Therefore } (A \cap B)^c = A^c \cup B^c$$

$$\text{Now } A \cup B = U$$

$$(A \cup B)^c = U^c = \emptyset$$

$$A^c \cap B^c = B \cap A = \emptyset$$

$$\text{Therefore } (A \cup B)^c = A^c \cap B^c$$

Self Assessment Questions

1. How do you represent the set of all students who have offered the MAHE course ?
2. Explain with proper reason, whether or not
 - i) the collections of all good singers is a set
 - ii) the set of points on a line is finite
3. Write the set of all integers using the tabular method.
4. Give two proper subsets and two supersets of the set of vowels of the English alphabet.
5. Under what conditions can a $A \cap B = A \cup B$?
6. Under what conditions on A and B is $A \times B = B \times A$?

1.8 Summary

In this units we have covered the following points:

1. The definition of sets as well-defined collections.
2. Different methods of representing sets.
3. The definition of subsets and equality of sets.
4. The operations of complementation, intersection and union of sets.
5. The definition of Cartesian product.
6. The use of Venn diagrams to represent sets.
7. Statement of the distributive laws and De Morgan's laws.

1.9 Terminal Questions

1. Explain the different types of sets with examples.
2. Define Cartesian product of two sets

1.10 Answers

Self Assessment Questions

1. $A = \{x : x \text{ is a student of MAHE}\}$
2. i) The collection of all good singers is not a set because this collection is not well defined. The characteristic good cannot be measured objectively.
ii) The set of points on a line is not finite because infinitely many points make straight line.
3. $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
4. Set of vowels of English alphabet $V = \{a, e, i, o, u\}$. Two subsets of set V are $V_1 = \{a, e\}$ and $V_2 = \{i, o\}$. Two supersets of V are $V_3 = \{a, b, c, \dots, z\}$ and $V_4 = \{a, c, d, e, i, o, u, \dots, z\}$.
5. $A \cap B$ can be equal to $A \cup B$ if either $A \subseteq B$ or $B \subseteq A$.
6. $A \times B = B \times A$ if and only if $A = B$

Terminal Questions:

1. Refer Section 1.3
2. Refer Section 1.6

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